Do Self-Supervised Speech Models Encode Phonological Features Language-Independently? A Probing Study of wav2vec 2.0 Variants

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Abstract

Self-supervised speech models such as wav2vec 2.0 learn strong representations from raw audio and are known to encode substantial phonetic information. I study whether they encode phonological features (voicing, nasality, manner and place of articulation) as language-independent properties, and whether languages share this latent structure. I probe three wav2vec 2.0 variants, a monolingual base model and two multilingual XLSR models, on English, German and Spanish (FLEURS), training linear probes on phoneme-level representations obtained by MMS forced alignment, and I evaluate three hypotheses across models (H1), layers (H2) and features (H3). Phonological information peaks in the upper-middle layers (about 70 to 75% of network depth) and declines toward the top, consistently across models (H2). Multilingual models hold a modest but consistent cross-lingual advantage over the monolingual baseline (H1). Features differ sharply in transferability: nasality's cross-lingual gap is statistically indistinguishable from zero, while manner of articulation shows the largest and clearly significant gap (H3). Transfer between all language pairs is broadly symmetric. Underlying these results, replacing uniform frame assignment with forced alignment roughly doubled probe accuracy, which shows that accurate phoneme to frame alignment is essential for this analysis.

1 Introduction

Self-supervised speech models such as wav2vec 2.0 have become important foundations for speech processing because they learn strong representations directly from raw waveform audio. I'm working with three models here: the base wav2vec 2.0 model, and two multilingual variants, XLSR-53, which was trained on 53 languages, and XLS-R, a 300-million-parameter model trained on 128 languages. Prior probing work suggests that the embeddings of these models encode phonetic and phonological information. In this research I aim to understand how that information is organised in the model and across different languages.

Even though we know these models capture phonetic information, it's still not clear whether they encode phonological features as language-independent concepts, or whether they just learn each language on its own. It's also unclear whether different languages end up sharing the same underlying phonological structure inside the model. These are the gaps I'm looking into.

This matters for a couple of reasons. First, it helps us understand what these black-box speech models are actually learning about speech sounds. Second, if phonology is encoded in a universal way, then a model trained on one language should transfer to another, which is useful for low-resource languages that don't have a lot of data. It also tells us whether multilingual pretraining is really worth it when the task is phonological.

I'm trying to answer these questions with three hypotheses:

- **H1:** multilingual models encode phonology more language-independently than monolingual ones, and therefore they have better cross-lingual transfer.
- **H2:** the middle layers of the model tend to encode the phonological information.
- **H3:** some of these features (voicing, nasality, place of articulation, and manner of articulation) transfer well across languages, while others are more language-specific.

To answer these questions, I extract the hidden representations from every layer of the three models, for English, German and Spanish speech from the FLEURS dataset. I then align each phoneme to its audio frames using forced alignment, and train a simple linear probe to predict each phonological feature from the representation. I check how well the probe transfers across languages, and I use cross-validation and confidence intervals so the results are statistically grounded and not just single numbers.

2 Related Work

Self-supervised speech models like wav2vec 2.0 (Baevski et al., 2020) learn representations from raw audio without labels, and they have become a common starting point for speech tasks. The multilingual versions, XLSR-53 (Conneau et al., 2021) and XLS-R (Babu et al., 2022), train on many languages at once and transfer well across them, at least for speech recognition.

Probing is a common way to check what a neural representation has actually learned. The idea is to train a simple classifier on top of a frozen model and see if it can recover some property of interest. Hewitt and Liang (2019) point out that a probe can sometimes memorize rather than reveal real structure, and they suggest control tasks to check for this.

For speech models specifically, earlier work has shown that wav2vec 2.0 embeddings contain a lot of phonetic and phonological information. Pasad et al. (2021) did a layer-wise analysis and found that this information is not spread evenly through the network, and that the layers follow an autoencoder-like pattern where the middle layers carry the most phonetic detail.

Most of this work looks at a single language, or at phonetic information in general. What I add here is a direct comparison between a monolingual and two multilingual models, a per-feature look at how well phonology transfers across languages, and confidence intervals so the comparisons are backed by statistics and not just single numbers.

3 Background

3.1 wav2vec 2.0

wav2vec 2.0 learns from raw audio without any labels. It has two parts: a convolutional feature encoder that turns the waveform into a sequence of frames, and a Transformer on top that adds context across those frames. It is trained by masking parts of the audio and learning to predict them, in a contrastive way where the model has to pick the correct piece out of a set of distractors. Because the training only uses the audio itself and no transcripts, the model has to discover useful structure on its own, and this is where the phonetic information is thought to come from. The output is one vector per frame at each layer, with about 50 frames per second. The multilingual variants I use, XLSR-53 and XLS-R, work the same way but are trained on many languages at once, which is meant to give them representations that work across languages.

feature	classes	examples
voicing	voiced, voiceless	voiced: b, d, z voiceless: p, t, s
nasal	nasal, oral	nasal: m, n, ng oral: all other sounds
manner	plosive, fricative, affricate, nasal, approximant	p (plosive), s (fricative), ch (affricate), m (nasal), l (approximant)
place	labial, coronal, dorsal, laryngeal	p (labial), t (coronal), k (dorsal), h (laryngeal)

Table 1: The four phonological features, their classes, and example sounds. Each feature is only probed on the segments where it is contrastive: voicing on obstruents, manner and place on consonants, and nasal on all segments.

3.2 Phonological features

Phonological features describe how a speech sound is physically produced, rather than what it means or how it is spelled. I look at four of them. Voicing is whether the vocal cords vibrate, which separates sounds like b, d, z from p, t, s. Nasality is whether air comes out through the nose, as in m, n, and ng. Manner of articulation is how the airflow is obstructed, going from a full block (plosive) to a narrow hiss (fricative) to almost no constriction (approximant). Place of articulation is where in the mouth the constriction happens, from the lips to the tongue tip to the back of the tongue to the throat. Table 1 lists them with examples. The important thing for this project is that these features are defined the same way in every language, so a nasal is a nasal whether the language is English, German, or Spanish. That is exactly what makes them a good test of whether the model has learned phonology in a language-independent way.

3.3 Linear probing

To measure whether the model encodes a feature, I train a simple linear classifier (a probe) on top of the frozen representation and check if it can predict the feature. If a linear probe can recover the feature, then the model must have encoded it in a clear, linearly readable way. I keep the probe linear on purpose, because a more powerful probe could find structure that the model did not actually expose, which would make the result harder to interpret. Figure 1 shows the idea: each point is a phoneme's representation, and the probe learns a boundary that separates the classes.

3.4 Evaluating a probe

The labels are not balanced. For voicing, for example, there are more voiceless obstruents than voiced ones, so a classifier that always guesses the majority class would already get a decent accuracy without learning anything. To avoid being fooled by this, I report macro-F1, which is the F1 score averaged over the classes so that each class counts equally, no matter how rare it is. I always compare the probe against a majority-class baseline, which is what you would get by always predicting the most common label, and I compute that baseline as macro-F1 too so it is on the same scale as the probe. A probe is only interesting if it clearly beats this baseline.

One more thing matters for a fair evaluation. Phonemes that come from the same recording are not independent, since they share the same speaker and the same audio. If I split the phonemes randomly, some phonemes from a recording could end up in training and others in testing, and the probe could get a good score just by recognizing the recording rather than the phonology. To

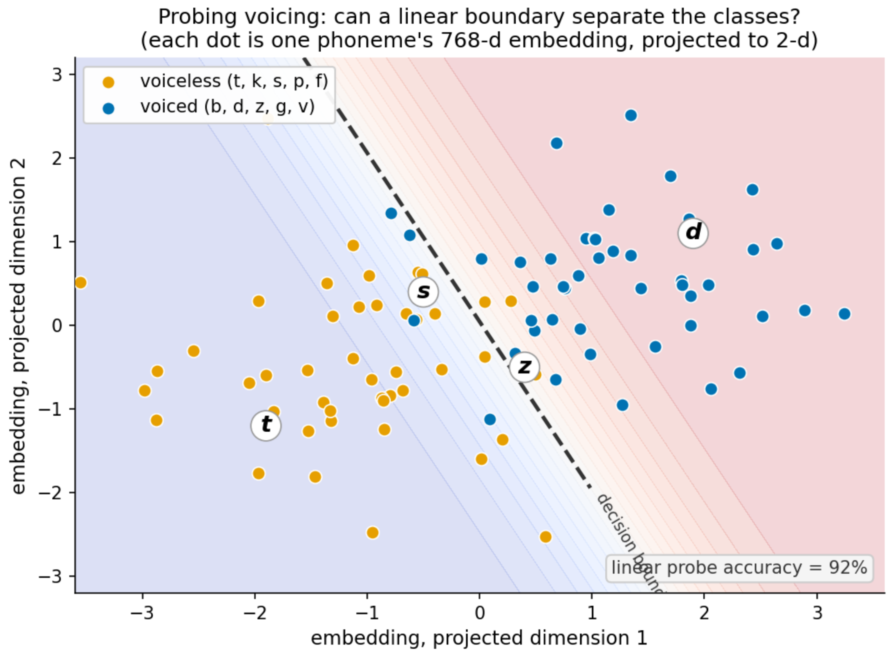

Figure 1: What a linear probe does, illustrated for voicing in two dimensions (the real representation has 768 dimensions). Each point is one phoneme; the probe learns a straight boundary that separates the classes. If the classes are linearly separable, the feature is encoded in a readable way.

prevent this I group the split by utterance, so all the phonemes from one recording stay on the same side of the split.

3.5 Forced alignment

The model produces frames, but the labels are phonemes, so I need to know which frames belong to which phoneme. Forced alignment solves this: given the audio and the transcript, which I already have, it finds where each word occurs in time. This is different from speech recognition, which tries to guess the words. Here the words are already known, and I only need their timing.

4 Methods

The pipeline has four steps, shown in Figure 2. I extract the hidden states from the models, turn the transcriptions into phonological labels, align the labels to the representations and train the probes, and then analyse the scores to answer the three hypotheses. The rest of this section describes each step.

4.1 Data

I use the FLEURS dataset, which is a multilingual read-speech corpus. I work with three languages: English (en_us), German (de_de), and Spanish (es_419), and I use 100 utterances per language at 16 kHz. FLEURS is parallel, which means the same sentences are translated across all the languages, so the three languages cover the same content and there's no topic bias between them. The actual

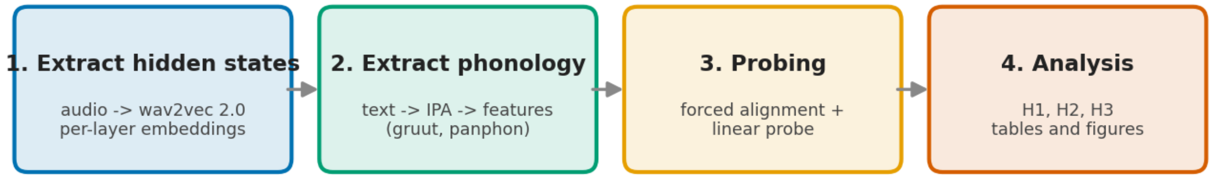

Figure 2: The four steps of the pipeline: extracting hidden states from the models, extracting phonological labels from the text, probing (which includes forced alignment and the linear probe), and the final analysis.

feature	en_us	de_de	es_419	total
voicing (obstruents)	3,225	4,378	3,895	11,498
nasal (all segments)	8,766	11,952	11,564	32,282
manner (consonants)	5,220	7,137	6,527	18,884
place (consonants)	5,220	7,137	6,527	18,884

Table 2: Number of phonemes used to probe each feature, per language, after the segment filter is applied. Nasal is probed on all segments, voicing only on obstruents, and manner and place only on consonants.

words and phonemes are still different in each language though, since they are translations of each other.

From these utterances I get 8,766 phonemes for English, 11,952 for German, and 11,564 for Spanish. The number of phonemes each probe actually uses depends on its segment filter, since each feature is only probed on the segments where it is contrastive. Table 2 shows how many phonemes are kept for each feature and language. Nasal uses every phoneme, so it has the most data, while voicing uses only obstruents and so has the least. The classes are also not balanced, for example there are more voiceless than voiced obstruents and far more oral than nasal segments, which is exactly why I use macro-F1 and a majority baseline rather than plain accuracy. The affricate class in manner is especially rare (fewer than 200 phonemes in total), which makes manner the hardest feature to probe.

4.2 Models

I compare three wav2vec 2.0 variants. The base model is monolingual and trained only on English, with 12 transformer layers and a hidden size of 768. XLSR-53 is multilingual, trained on 53 languages, with 24 layers and a hidden size of 1024. XLS-R is also multilingual, trained on 128 languages, with the same 24 layers and hidden size of 1024. The base model is my monolingual baseline for the H1 comparison.

4.3 Extracting hidden states

I run each model on the utterances without any fine-tuning, so the models stay frozen. For every utterance I keep the output of every layer, not just the last one, because H2 is about finding which layer encodes the phonological information. Each layer gives a grid of shape (T, D), where T is the number of time frames (about 50 per second, so one frame every 20 ms) and D is the hidden size (768 or 1024).

4.4 Extracting phonology

To get the labels, I convert each transcription into phonemes in two steps. First, gruut does the grapheme-to-phoneme conversion, turning the text into IPA. Then panphon (Mortensen et al., 2016) maps each IPA symbol to its distinctive features. I collapse those into four labels: voicing (voiced or voiceless), nasal (nasal or oral), manner (plosive, fricative, affricate, nasal, or approximant), and place (labial, coronal, dorsal, or laryngeal).

Each feature is only probed on the segments where it actually means something. Voicing is probed on obstruents only, because sonorants are almost always voiced, so probing it on everything would give a nearly constant label. Manner and place are probed on consonants only, since they are undefined for vowels. Nasal is probed on all segments.

4.5 Aligning phonemes to frames

To attach the labels to the representations, I need to know which audio frames belong to which phoneme. I use the MMS forced aligner (Pratap et al., 2024) to get word-level time spans, which tell me where each word occurs in the audio. I then split each word's time span evenly across its phonemes to get a time span for each phoneme. Each phoneme's time span is converted into frame indices using the clip duration, and I mean-pool the frames inside that span into a single vector. This gives me one representation per phoneme, paired with its label. This replaced an earlier version that just split the whole utterance evenly across all its phonemes, which was a lot less accurate.

4.6 Probing

For each combination of model, feature, layer, and language, I train a linear probe (logistic regression) to predict the label from the phoneme vector. I standardize the features first. I keep the probe linear on purpose, because that tests whether the feature is linearly readable from the representation, so the probe can't invent structure that the model didn't actually expose. I report macro-F1, since the labels are imbalanced and plain accuracy would be misleading, and I compare it against a majority-class baseline that is also computed as macro-F1. The train and test splits are grouped by utterance, so phonemes from the same recording never end up on both sides, which avoids leakage. I repeat this over 5 grouped splits and report the mean and standard deviation.

4.7 Cross-lingual transfer and statistics

For the cross-lingual part, I do zero-shot transfer: I train the probe on one language and test it on the others, covering all the ordered language pairs. I report transfer at each model and feature's best layer, chosen by within-language macro-F1, so I'm not picking an arbitrary layer.

For H3, I measure the transfer gap, which is the within-language macro-F1 minus the cross-lingual macro-F1. I use a paired k-fold procedure: I split language A's utterances into 5 folds, repeated 3 times, and in each fold I train one probe and test it both on the held-out utterances of A (within) and on the other languages (cross). Because both scores come from the same probe and training set, the gap is paired. This gives 45 gap estimates per model and feature, and I report the mean and a 95% confidence interval from the percentiles. I treat a gap as real if its confidence interval excludes zero. I also compute the paired differences between features on the same folds. I

feature	uniform alignment	forced alignment	change
voicing	0.54	0.85	+0.31
nasal	0.47	0.76	+0.29
place	0.34	0.77	+0.43
manner	0.22	0.67	+0.45

Table 3: Effect of alignment on within-language decodability (best within-language macro-F1, base model). Switching from uniform to MMS forced alignment roughly doubled every feature, using the same models and probes.

use a percentile confidence interval rather than a standard t-test, because the folds share most of their training data and so their scores are correlated, which would make a t-test overconfident.

5 Results

The results in this section improved a lot over the course of the project, and it is worth showing this because it makes clear how much the method matters. My first complete run used uniform alignment, splitting each utterance's frames evenly across its phonemes. The scores were weak and flat: within-language voicing only reached about 0.54 and manner about 0.22, barely above their baselines, so it looked like the models were not encoding much. Replacing uniform splitting with MMS forced alignment was the turning point. Lining each phoneme up with the frames it actually occupies roughly doubled the scores, for example within-language voicing went from about 0.54 to 0.85 and manner from about 0.22 to 0.67 (Table 3). The models and the probes were exactly the same, so the improvement came entirely from better alignment, which showed that the earlier weak results were a measurement problem and not a real absence of phonological information. A final round of work on grouped splits, error bars, best-layer selection and the paired k-fold test did not raise the numbers, but it made them trustworthy. All the results below use forced alignment.

5.1 H2: which layers encode phonology

Figure 3 shows the macro-F1 of each feature at each layer. All four features are well above their baselines at every layer, so the phonological information is decodable through the whole network. But the curves are not flat. They start lower at the input, rise up, peak in the upper-middle layers, and then drop off again toward the top. For the base model the peak is around layers 8 to 10 out of 12, and for the two larger models it is around layers 16 to 18 out of 24. In both cases that is roughly 70 to 75% of the way through the network, and this is consistent across all three models. The features also keep the same order at almost every layer: voicing is the easiest to decode, place and nasal sit close together in the middle, and manner is the hardest. The drop in the top layers makes sense, since those layers tend to specialize back towards the model's own pretraining objective. Overall this supports H2.

5.2 H1: multilingual vs monolingual

Table 4 shows the cross-lingual transfer for each model, measured at each feature's best layer. The two multilingual models match or beat the monolingual base model on every feature. The biggest differences are on voicing, where base gets 0.67 and XLS-R gets 0.75, and on manner, where base gets 0.45 and XLS-R gets 0.53. But most of these differences are small, often around the same

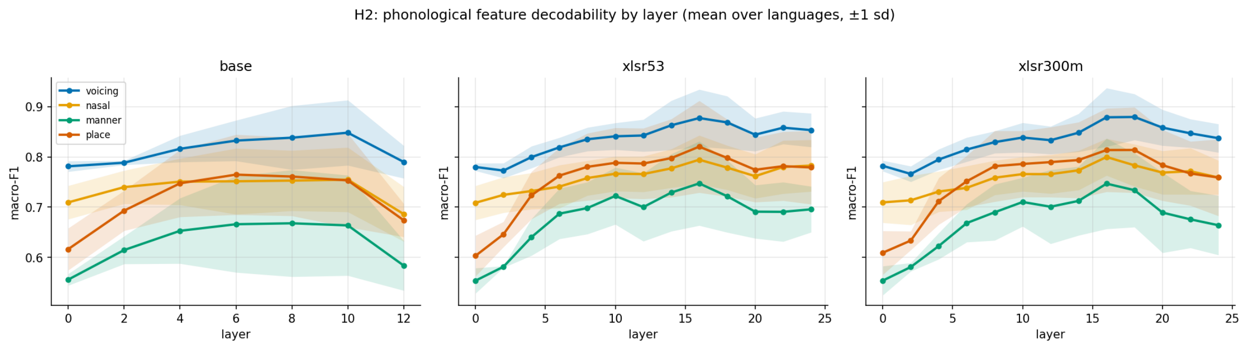

Figure 3: H2: macro-F1 by layer for each model (mean over languages, shaded band is ±1 standard deviation across languages). All features peak in the upper-middle layers and decline toward the top.

feature	base	XLSR-53	XLS-R	majority baseline
voicing	0.667	0.721	**0.753**	0.38
nasal	0.720	**0.760**	0.728	0.47
place	0.628	**0.660**	0.658	0.26
manner	0.452	0.513	**0.533**	0.09

Table 4: H1: cross-lingual transfer macro-F1 (all language pairs, at the best layer). Bootstrap standard deviations are about 0.04.

size as the bootstrap standard deviation of about 0.04, so the advantage is consistent in direction but not very large. So H1 is only partly supported: for these three related languages, multilingual pretraining does give a cross-lingual benefit, but a modest one.

Figure 4 shows the same comparison as a bar chart, which makes the small size of the differences easier to see. The multilingual models are usually a little ahead, but the bars are close and the error bars overlap for most features, which is why I describe the advantage as consistent but modest.

I also checked whether the model differences depend on the layer, by looking at cross-lingual transfer at every layer rather than only the best one (Figure 5). The models track each other closely at every depth, so the modest multilingual advantage is not something that only appears at one lucky layer, it holds across the network.

5.3 H3: which features transfer across languages

Figure 6 shows the transfer gap for each feature with its 95% confidence interval, where the gap is the within-language score minus the cross-lingual score. A gap is only real if its confidence interval stays above zero. Manner has the largest gap in every model, around 0.21 to 0.24, and it is clearly significant, so it is the most language-specific feature. Nasal is the opposite: its gap is small (0.03 to 0.07) and its confidence interval includes zero in all three models, so its gap is statistically indistinguishable from zero. In other words nasality transfers across languages essentially for free. Voicing is significant in all three models, and place is significant in two of them and borderline in the third. The important part is that the ranking, manner then voicing and place then nasal, is identical across all three models, which suggests it reflects a property of phonology itself and not just one model. The paired test also confirms that the difference between manner and nasal is significant in every model. This supports the core of H3, but not my original guess that place

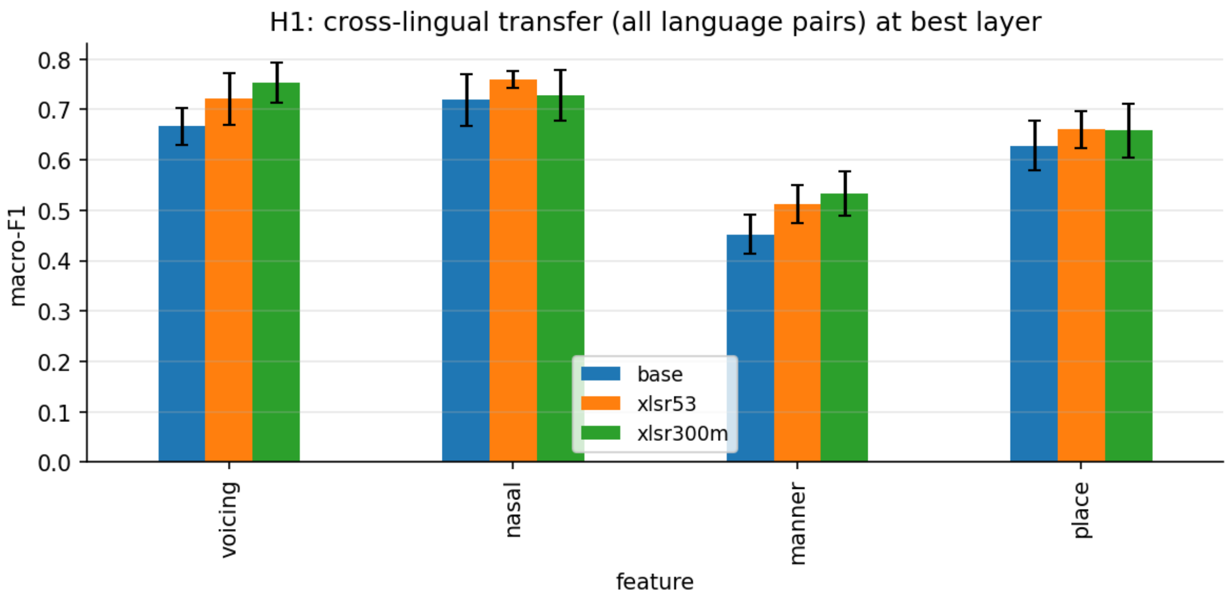

Figure 4: H1: cross-lingual transfer macro-F1 per feature, grouped by model, at the best layer. The multilingual models are usually slightly ahead of base, but the differences are small.

would be the most language-specific feature, since that turned out to be manner, and nasal turned out to be the most universal.

The same result can be seen in a more intuitive way in Figure 7, which plots the within-language and cross-lingual scores side by side with a band for their spread across the k-fold splits. When the two bands overlap, the gap is within the noise, and when they are clearly apart, the gap is real. For nasal the two bands sit almost on top of each other, while for manner they are far apart. This is the same conclusion as the confidence intervals, just shown visually.

Finally, I tested whether the features are significantly different from each other, not just from zero. Figure 8 shows the paired differences between every pair of features with their confidence intervals. The manner versus nasal difference is clearly significant in every model, which confirms that manner really is more language-specific than nasal and this is not just noise.

Table 5 collects the full numbers behind these figures: for every model and feature it gives the best layer, the within-language and cross-lingual scores, the transfer gap, its 95% confidence interval, and whether the gap is significant. The pattern is consistent across models. Nasal is never significant, manner and voicing are always significant, and place is significant for two of the three models and just borderline for the third (its lower bound is almost exactly zero).

5.4 All-pairs transfer

I also looked at transfer between every pair of languages (Figure 9). The matrix is roughly symmetric, so training on any one of the three languages and testing on the others gives similar scores. This fits the idea that the three languages share a lot of the same latent phonological structure.

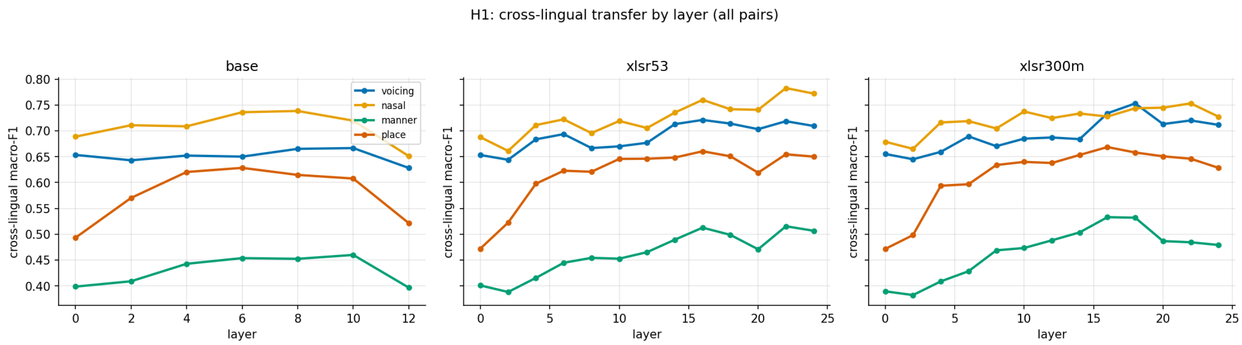

Figure 5: Cross-lingual transfer by layer for each model. The models follow similar curves at every depth, with the multilingual models usually slightly ahead, so the advantage is not tied to any particular layer.

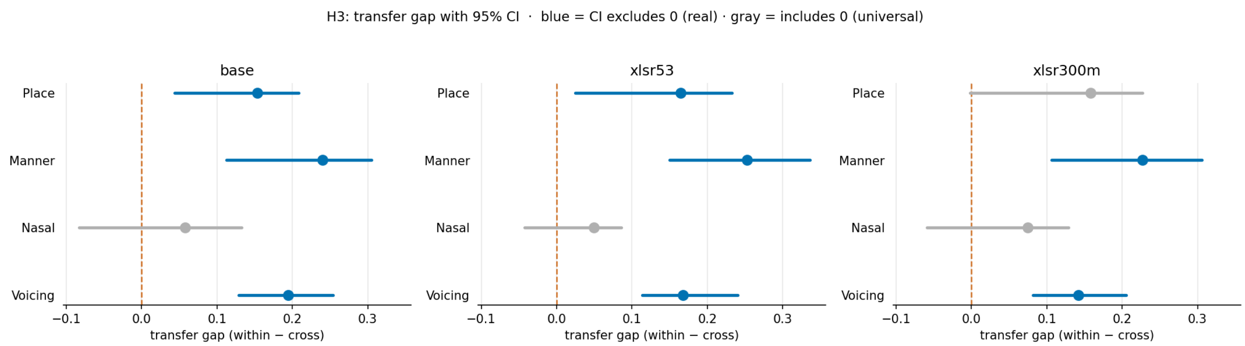

Figure 6: H3: transfer gap (within-language minus cross-lingual macro-F1) with 95% confidence intervals. Blue means the interval excludes zero (a real gap); grey means it includes zero (indistinguishable from language-independent). Nasal is grey in all models; manner shows the largest gap.

6 Discussion

Alignment turned out to be the most important part. The single biggest thing I learned is how much the phoneme to frame alignment matters. My earlier version just split each utterance's frames evenly across all its phonemes, which ignored the leading silence and the fact that phonemes have very different lengths. With that approach the probes were barely above chance. Once I switched to MMS forced alignment the scores roughly doubled, for example within-language voicing went from about 0.54 to about 0.85. So phoneme-level probing of these models is very sensitive to alignment quality, and a bad alignment can completely hide the structure that is actually there.

What the results mean. The layer pattern in H2 fits the usual view that these models build up more abstract phonetic information in the middle and then specialize the top layers back towards their pretraining task. This also matches what Pasad et al. (2021) found in their layer-wise analysis, where the middle layers carried the most phonetic detail, so it is reassuring to see the same shape here for phonological features and across three languages rather than just one.

The modest multilingual advantage in H1 probably has a lot to do with the languages I used. English, German and Spanish are all Indo-European and share most of their sounds, so even a monolingual English model has already seen most of the phonemes it needs, and there is not much

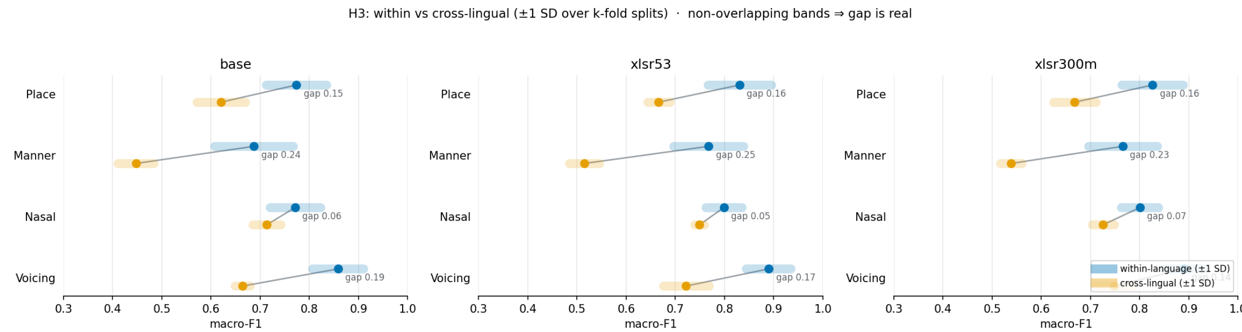

Figure 7: H3 shown as within-language versus cross-lingual scores, with bands for the spread across k-fold splits. Overlapping bands mean the gap is not real (nasal); clearly separated bands mean it is (manner).

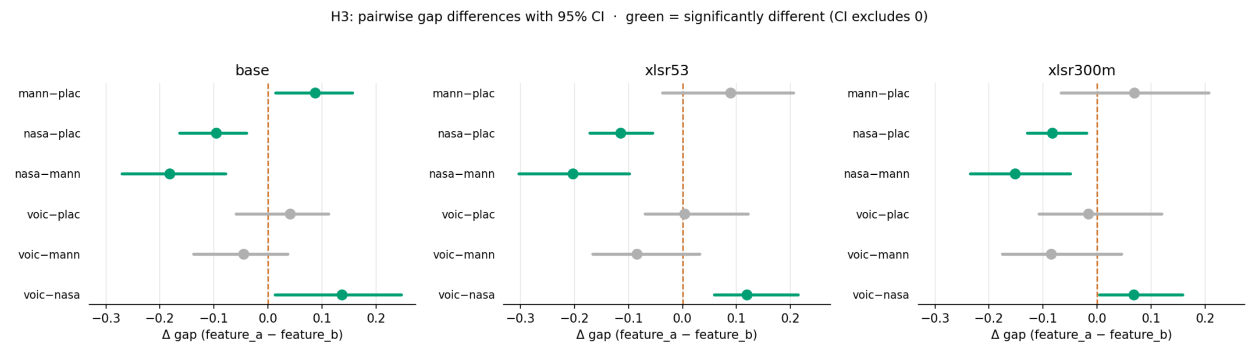

Figure 8: H3: paired differences between feature gaps, with 95% confidence intervals. Green means the two features differ significantly (interval excludes zero). The manner versus nasal difference is significant in all models.

room left for multilingual pretraining to help. A bigger advantage might show up for languages that are more different from English. The feature ranking in H3 makes linguistic sense too: nasality is acoustically strong and realized in a similar way across languages, while manner covers five categories with finer differences that vary more between languages.

Limitations. The main limitation is the language sample. I only used three languages, and they are all Indo-European with a lot of overlap in their sounds. So my statistical tests are valid within English, German and Spanish, and I can say things like manner transfers worse than nasal in these languages, but I cannot claim that any feature is universal across all human languages, because for that kind of claim the real sample size is just three. There are also two smaller limitations. The alignment is only accurate at the word level, and inside each word I still split the phonemes evenly, so a phoneme-level aligner would remove that last approximation. And the best layer for H1 and H3 is chosen on the same data I report on, though I reduce this problem for H3 by measuring the gap on held-out folds.

Future work. The most useful next step would be to add languages that are very different from English, like a tonal language or a click language, because that is the real test of whether these features are universal. A control task (Hewitt and Liang, 2019) would also help, by checking how

model	feature	best layer	within	cross	gap [95% CI]	significant
base	voicing	10	0.848	0.667	0.182 [0.128, 0.254]	yes
base	nasal	10	0.755	0.720	0.035 [−0.083, 0.132]	no
base	place	6	0.765	0.628	0.136 [0.044, 0.209]	yes
base	manner	8	0.668	0.452	0.215 [0.113, 0.305]	yes
XLSR-53	voicing	16	0.878	0.721	0.156 [0.114, 0.241]	yes
XLSR-53	nasal	16	0.795	0.760	0.035 [−0.043, 0.086]	no
XLSR-53	place	16	0.821	0.660	0.160 [0.025, 0.232]	yes
XLSR-53	manner	16	0.747	0.513	0.235 [0.150, 0.336]	yes
XLS-R	voicing	18	0.880	0.753	0.127 [0.081, 0.205]	yes
XLS-R	nasal	16	0.800	0.728	0.072 [−0.059, 0.129]	no
XLS-R	place	18	0.814	0.658	0.156 [−0.002, 0.227]	no
XLS-R	manner	16	0.747	0.533	0.214 [0.106, 0.306]	yes

Table 5: Full H3 results per model and feature: best layer, within-language and cross-lingual macro-F1, the transfer gap with its 95% confidence interval, and whether the gap is significant (interval excludes zero). Within-language scores are also a useful summary of how decodable each feature is.

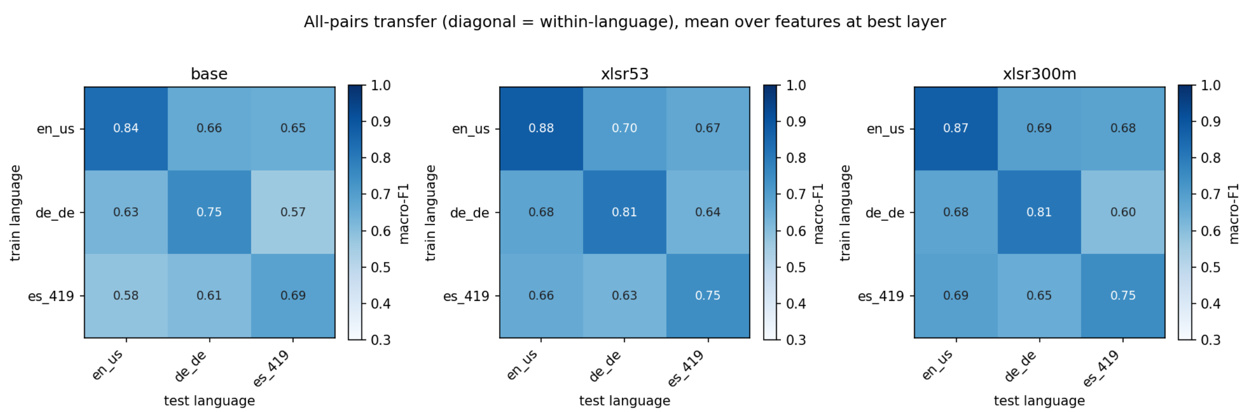

Figure 9: All-pairs transfer matrix per model (rows are the training language, columns are the test language, averaged over features at the best layer). The diagonal is within-language performance.

much of the probe's score is real signal versus memorization. Also futhermore fine-tuning the models and repeating the probing would be a natural extension, especially for H1.

7 Conclusion

I set out to study whether wav2vec 2.0 and its multilingual variants encode phonological features as language-independent properties. Using linear probes on phoneme-level representations from English, German and Spanish, I found that phonological information is concentrated in the upper-middle layers and drops off toward the top, and this is consistent across all three models (H2). The multilingual models are usually a little better than the monolingual base model at cross-lingual transfer, but the advantage is modest for these related languages (H1). The features differ clearly in how well they transfer: nasality's transfer gap is statistically indistinguishable from zero, so it behaves as if it is language-independent, while manner of articulation has the largest gap and is the most language-specific (H3). The same ranking of features shows up in all three models, which suggests it reflects phonology itself rather than any one model.

The clearest practical lesson is about method rather than models. Getting the phoneme to frame alignment right was what made the difference between results that looked like noise and results that were clearly meaningful, since forced alignment roughly doubled the probe scores. Beyond the specific findings, this is a reminder that for phoneme-level probing of speech models, the alignment step deserves as much care as the probing itself.

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A Implementation details

All experiments were run in Python. The models are loaded from the Hugging Face Transformers library, and the FLEURS data is loaded from the Hugging Face Datasets library. The three models are `wav2vec2-base`, `wav2vec2-large-xlsr-53` and `wav2vec2-xls-r-300m`. The audio is kept at 16 kHz, which is what the models expect.

For the phonology, I use gruut for grapheme-to-phoneme conversion and panphon for the distinctive features. Because these are slow, I run them once for all the utterances and cache the results.

The forced alignment uses the MMS aligner from torchaudio. MMS expects text in a romanized form using only the letters a to z, but gruut needs the original spelling to phonemize German and Spanish correctly. I handle this by keeping two versions of each word: the original spelling, with its accents, is used for phonemization, and a romanized version is used only for the aligner. The alignment for all utterances is also computed once and cached.

The probe is a logistic regression from scikit-learn, with a standard scaler in front of it, a maximum of 2000 iterations, and balanced class weights. Each within-language score is the mean over 5 repeated grouped splits, where the split is grouped by utterance. For the H3 gap test I use 5 folds repeated 3 times for each of the 3 training languages, which gives 45 gap estimates per model and feature, and I take the 95% confidence interval from the 2.5th and 97.5th percentiles.

B Per-language transfer

Table 6 gives the full all-pairs transfer for each model, averaged over features at the best layer. The rows are the training language and the columns are the test language, so the diagonal is within-language performance and the off-diagonal entries are cross-lingual transfer. The matrices are close to symmetric, and the off-diagonal values are fairly similar regardless of which language is used for training, which supports the idea that the three languages share a lot of the same phonological structure.

model	train \ test	en_us	de_de	es_419
base	en_us	0.835	0.661	0.651
base	de_de	0.629	0.754	0.565
base	es_419	0.583	0.611	0.687
XLSR-53	en_us	0.878	0.703	0.670
XLSR-53	de_de	0.678	0.806	0.637
XLSR-53	es_419	0.659	0.634	0.746
XLS-R	en_us	0.873	0.693	0.684
XLS-R	de_de	0.679	0.808	0.604
XLS-R	es_419	0.693	0.653	0.749

Table 6: All-pairs transfer macro-F1 for each model (rows are the training language, columns are the test language), averaged over features at the best layer. The diagonal is within-language performance.